

1 Solution — GIAM 3.1

Problem 5

1.1 Problem

Prove that if the sum of two integers is even, then so is their difference.

1.2 Statement

$$\forall x, y \in \mathbb{Z}, \quad (x + y \text{ is even}) \implies (x - y \text{ is even}).$$

1.3 Proof

Proof. Let x and y be integers and suppose $x + y$ is even. By the definition of even, there is an integer k with $x + y = 2k$. By algebra,

$$x - y = (x + y) - 2y = 2k - 2y = 2(k - y).$$

Now $k - y$ is an integer, since $\mathbb{Z}$ is closed under subtraction. So $x - y = 2m$ with $m = k - y \in \mathbb{Z}$, and hence $x - y$ is even. ■

The whole proof rests on the identity $x - y = (x + y) - 2y$, which converts the *hypothesis* directly into the *conclusion*.

1.4 Why No Case Analysis Is Needed

One algebraic identity does the whole job — no case analysis required.

The one-identity proof

assume $x + y$ is even: $x + y = 2k$ for some $k \in \mathbb{Z}$

$$x - y = (x + y) - 2y = 2k - 2y = 2(k - y)$$

and $k - y \in \mathbb{Z}$, so $x - y = 2m$ with $m = k - y \in \mathbb{Z}$

hence $x - y$ is even ■

Why no case analysis is needed

$$(x + y) - (x - y) = 2y \text{ and } (x + y) + (x - y) = 2x$$

Both are even, so $x+y$ and $x-y$ always have the same parity — whatever x and y are.

The parity table confirms it: the two columns are identical

x	y	$x + y$	$x - y$	same?
even	even	even	even	✓
even	odd	odd	odd	✓
odd	even	odd	odd	✓
odd	odd	even	even	✓

Shaded rows are the ones where the hypothesis holds — and there the conclusion holds too.

The converse holds too, so the real theorem is a biconditional:

$$x + y \text{ even} \iff x - y \text{ even} \iff x \text{ and } y \text{ have the same parity}$$

Figure 1.1: Top panel, the one-identity proof: assume x plus y is even, so x plus y equals $2k$ for some integer k ; then x minus y equals $(x$ plus $y)$ minus $2y$ equals $2k$ minus $2y$ equals 2 times $(k$ minus $y)$; and k minus y is an integer, so x minus y equals $2m$ with m an integer, hence x minus y is even. Second panel, why no case analysis is needed: $(x$ plus $y)$ minus $(x$ minus $y)$ equals $2y$, and $(x$ plus $y)$ plus $(x$ minus $y)$ equals $2x$ — both are even, so x plus y and x minus y always have the same parity, whatever x and y are. Below, a four-row parity table with columns x , y , x plus y , x minus y , and ‘same?’. Row even-even: sum even, difference even, same. Row even-odd: sum odd, difference odd, same. Row odd-even: sum odd, difference odd, same. Row odd-odd: sum even, difference even, same. The two shaded rows are those where the hypothesis holds, and there the conclusion holds too. A closing panel notes the converse holds as well, so the real theorem is a biconditional: x plus y even if and only if x minus y even if and only if x and y have the same parity.

A tempting first approach is to split into cases — “both x and y even” and “both odd” — since those are the two ways a sum can be even. That works, but it is unnecessary. The reason is the pair of identities

$$(x + y) - (x - y) = 2y, \quad (x + y) + (x - y) = 2x.$$

Both right-hand sides are even, so $x + y$ and $x - y$ always differ by an even number — meaning they always have the *same parity*, regardless of x and y . The theorem is

then immediate. Checked exhaustively on $[-40, 40]^2$: no counterexamples.

1.5 Remark — The Case Proof Also Works

For comparison, here is the case-analysis version, which is perfectly correct but longer.

Since $x + y$ is even, x and y have the same parity (a fact needing its own justification — see Problem 4's table).

- **Both even:** $x = 2a$, $y = 2b$, so $x - y = 2a - 2b = 2(a - b)$, even.
- **Both odd:** $x = 2a + 1$, $y = 2b + 1$, so $x - y = (2a + 1) - (2b + 1) = 2(a - b)$, even.

Both branches conclude “even,” so the result follows by **proof by cases** — the constructive-dilemma pattern of Problem 2.6.2. Notice, though, that this version needs an extra lemma (“even sum implies matching parity”) that the one-identity proof never touches. Preferring the shorter route is not just aesthetics: fewer steps means fewer places to make a mistake, and no auxiliary result to prove first.

1.6 Remark — The Converse Is Also True

The hypothesis and conclusion are interchangeable here. Running the same identity backwards, $x + y = (x - y) + 2y$, shows that if $x - y$ is even then so is $x + y$. So the honest theorem is a **biconditional**:

$$x + y \text{ is even} \iff x - y \text{ is even} \iff x \text{ and } y \text{ have the same parity.}$$

Verified exhaustively: neither direction has a counterexample on $[-40, 40]^2$. This is worth flagging because, as Problem 2.2.9 stressed, a converse normally has to be *earned* — most conditionals are one-way. Here it comes essentially free, since the proof was a reversible algebraic rearrangement rather than a one-directional deduction.

1.7 Remark — Where the Hypothesis Is Actually Used

The hypothesis “ $x + y$ is even” enters at exactly one point: it supplies the integer k with $x + y = 2k$. Everything after that is algebra plus closure. This is a good habit to check

on any proof — if you cannot point to the line where a hypothesis was used, either the hypothesis is unnecessary or the proof is wrong.

Here the hypothesis is genuinely needed. Without it there is no reason for $x - y$ to be even: take $x = 5$, $y = 2$, where $x + y = 7$ is odd and $x - y = 3$ is odd too. The theorem says nothing about that case, and correctly so.

1.8 Remark — The General Pattern: Two Expressions Differing by an Even Amount

The technique generalises well beyond this problem. To show two quantities share a parity, it suffices to show their **difference is even** — you never need to determine either parity individually. Some instances:

claim	difference	why even
$x + y$ and $x - y$	$2y$	visibly a multiple of 2
n and $n + 2$	2	consecutive same-parity integers
n and n^2	$n^2 - n = n(n - 1)$	product of consecutive integers
n and n^3	$n^3 - n = (n - 1)n(n + 1)$	product of three consecutive integers

Table 1.1

The last two are the reason n , n^2 , and n^3 always share a parity — a fact used constantly, and proved each time by the same “their difference is even” move rather than by splitting into cases.